Cume February 19, 2016 - Spectroscopy

Spectroscopy is arguably the most important tool for investigating the structure of matter, and is also the basis of a number of important analytical methods. All spectroscopies involve the interaction of electromagnetic radiation with matter. Depending on the technique, a photon can be absorbed, a photon can be emitted, or a photon can be absorbed and another photon emitted. Because energy is conserved, the net photon energy stays in the molecule and changes its internal state. This can be a change in the electronic wavefunction, in which an electron is transferred from one orbital to another, a change in the vibrational wavefunction where the state of the nuclear vibrations change, a change in the rotational wavefunction (for freely rotating gas-phase molecules only), a change in the spin direction of the electrons, or a change in the spin direction of the nuclei. (Often several of these change simultaneously but the most important is the one that is associated with the largest energy change).

(I) For each of the spectroscopic techniques described below, give

(a) A brief description of the technique (e.g. usually absorption, emission or absorption and emission)
(b) The main change in the internal status of the molecule
(c) A typical wavelength of the electromagnetic radiation used in this type of spectroscopy (in m or its standard multiples, e.g. μm or nm).
(d) The typical frequency associated with the wavelength in (c), in units of Hz.
(e) The wave number of the light, expressed in cm^{-1}
(f) The photon energy associated with the wavelength in (c), in units of J.

Constants: speed of light $c = 2.998\times10^{8}$ m/s, Planck's constant $h=6.626\times10^{-34}$ Js.
For your convenience, the first question in (I) is worked out. (70 %)

(II) Define the concept of *selection rule* (15 %)

(III) What is the selection rule for transitions between vibrational energy levels of a molecule if the vibrations are harmonic? (15 %)

Worked out answer for (I-1))
1. Visible spectroscopy. (a) It is most often measured in absorption, (b) It is associated with electronic transitions in molecules with low-lying electronic states,
(c) 0.5×10^{-6} m = 500 nm (d) 6.0×10^{14} Hz (e) 20000 cm^{-1} (f) 3.98×10^{-19} J = 0.398 aJ

2. Infrared spectroscopy.

3. Ultraviolet spectroscopy

4. NMR spectroscopy

5. Fluorescence spectroscopy

6. Microwave spectroscopy

7. X-ray photoelectron spectroscopy

8. Raman spectroscopy

(II) The concept of a selection rule. What is the underlying reason for selection rules? (15 %)

(III) The selection rule for the vibrational transitions of a molecule if the vibrations are harmonic